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Bioaccumulation and sub-chronic toxicity of microplastic environmentally relevant concentrations in *Etroplus suratensis* brackish water fish

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Abstract

Numerous ecosystem-based studies have explored microplastic pollution in brackish water environments. However, research on the bioaccumulation of microplastics in brackish water fish and their effects remains limited. The present study investigated the bioaccumulation and effects of polystyrene microplastics (PS-MPs) in juvenile brackish water fish, pearl spot (*Etroplus suratensis*). Fish were exposed to 0, 0.2, 2, and 4 mg/L of 1 µm-sized PS-MPs for 14 days. PS-MPs were found in the gastrointestinal tract, gills, liver, spleen, muscle, and brain, with the highest concentration in the intestine and the lowest in the brain. Exposure to PS-MPs led to elevated serum level of glucose, total protein, total cholesterol, serum glutamic-oxaloacetic transaminase (SGOT), serum glutamic-pyruvic transaminase (SGPT), and alkaline phosphatase (ALP). Antioxidant parameters such as superoxide dismutase (SOD), catalase (CAT), glutathione peroxidase (GPx), and total antioxidant capacity (TAC) decreased, while malondialdehyde (MDA) level and protein carbonyl (PC) content increased. PS-MP exposure down-regulated hepatic expression of *NRF2* and *P53*, increased cortisol levels, and up-regulated *HSP70* gene expression in a dose-dependent manner. Additionally, PS-MPs down-regulated expression of *IGF1* and *CYP1A* in the liver. This is the first comprehensive research that has revealed the extent to which PS-MPs accumulate in various tissues of brackish water fish species after being exposed to environmentally significant concentrations. It also demonstrates the associated toxicity in an array of antioxidant indicators.

Keywords Pearl spot · Polystyrene microplastic · Bioaccumulation · Toxicity · Antioxidant activity

Introduction

The accumulation of micro and nanoplastics (MPs and NPs) in diverse ecosystems poses a global menace and presents a formidable challenge to address. According to Statista (2023), the ocean is exposed to approximately 1.3 million tonnes of microplastics annually. Recently, the novel laser14 backscattered fiber-embedded optofluidic chip (LFOC) was developed as a more rapid and reliable technique for quantifying nanoplatics in aquatic environments, with a minimum detection limit of 0.23 µg/mL (Lu et al. 2024). Microplastics have been detected in numerous aquatic environments, thereby endangering aquatic organisms such as fish with potential exposure risks (Baldwin et al. 2016; Lusher et al. 2016; Dai et al. 2018). When fish consume plastic particles, such as MPs and NPs, these particles tend to accumulate in the digestive tract for a prolonged period. This accumulation can have harmful consequences for aquatic organisms, including cytotoxicity, genotoxicity, neurotoxicity, and reproductive toxicity (Das et al. 2023; Pang et al. 2021). However, particles exceeding 100 µm in size do not appear to induce adverse hepatic stress in fish (Ašmonaitė et al. 2018; Guerrera et al. 2021). Smaller MPs and NPs have a higher potential to cause harm as they can easily move through biological membranes. Both aquatic life and human health are at risk due to the bioaccumulation of smaller MPs and NPs pollutants by organisms at higher trophic levels through the food chain (Hu and Palic 2020; Guerrera et al. 2021). In each fish

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species, the response to MP exposure varies based on their functional traits (Salerno et al. 2021; Zhang et al. 2023).

In 2021, India produced approximately 130,000 tonnes of ocean plastic, marking the second-largest volume in Asia (Statista 2023). Furthermore, the presence of MPs has been documented across various regions of India, including both the east and west coasts, in sediment, water, zooplankton, fish, bivalves, salt, shrimp, benthic invertebrates, and atmospheric dust (Veerasingam et al. 2020). India's first report of MP pollution in lake sediments was from Vembanad lake in Kerala (Sruthy and Ramasamy 2017). MPs in Vembanad lake had an average abundance of 26.79 ± 3.74 items L^{-1} in subsurface waters and 52.70 ± 5.43 items L^{-1} in bottom waters (Anagha et al. 2023). This environmental MP concentration, equivalent to 0.2 mg/L, is selected as the lowest MP exposure concentration for this study. Several studies have revealed the presence of MPs in various organisms, including fish (James et al. 2020; Robin et al. 2020), shrimp gastrointestinal tracts (Daniel et al. 2020), and benthic invertebrates such as polychaetes and bivalves along the Kerala coast (Naidu et al. 2018). Additionally, research conducted in the Sal estuary of Goa found an average MP concentration of 1.4 ± 0.3 MPs per gram of body weight in *Etroplus suratensis* (Saha et al. 2021). Moreover, the average abundance of MPs from the digestive tracts of *Etroplus suratensis* caught from brackish water Vembanad Lake of India was reported to be 13 ± 5 nos./individual (Nikki et al. 2021). These findings underscore the ecosystem-based approach adopted by Indian studies to investigate microplastic pollution. However, thorough examinations of the potential adverse effects of MPs on fish health in India are still lacking.

The Asian cichlid fish species, pearl spot (*Etroplus suratensis*), is widely distributed in the India and Sri lanka (Rishi and Singh 1982). It is a commercially important fish species which constitutes a major fishery in brackish water lakes in India (Rattan 1994; Munilkumar et al. 2013). Additionally, it has a high tolerance for salinity and can be cultured in both freshwater and brackish water (Chandrasekar et al. 2014).

There are no sufficient detailed studies on brackish water fish related to MP bioaccumulation and its effects. Therefore, the primary objective of this study was to investigate the bioaccumulation and effects of MPs on *Etroplus suratensis*, an economically significant indigenous brackish water fish species highly susceptible to MP pollution in India. This research marks the first comprehensive examination of the bioaccumulation and toxicity of smaller MPs in juvenile pearl spot, focusing on environmentally relevant concentrations. We exposed the fish to environmentally relevant concentrations of microplastic at 0.2, 2, and 4 mg/L (Besseling et al. 2019; Abidli et al. 2021; Sun et al. 2021; Kye et al. 2023; Xiong et al. 2023). Through our findings, we aim to elucidate the underlying mechanisms governing the impact of smaller MP pollutants on fish health. Ultimately, this study lays a crucial foundation for future research endeavors aimed at safeguarding the sustainability of native fish populations in polluted aquatic environments.

Materials and methods

Characterization of the microplastic

Virgin PS-MPs (mean diameter = 1 µm, aqueous suspension with concentration 10% and density 1.05 g/cm^3) and yellow-green fluorescent PS-MPs (mean diameter 1 µm, aqueous suspension with concentration 2.5% and density 1.05 g/cm^3, λex = 470 nm and λem = 505 nm) were purchased from Sigma Aldrich, USA. To prepare the PS-MP suspension, deionized water was used and it was sonicated before every use to ensure thorough dispersion of MP particles in the water. The microplastic morphology and composition were analyzed using Scanning Electron Microscope (SEM) and Confocal Raman Microscope, respectively. For SEM examination, a scanning electron microscope (FE-SEM, Joel JSM 5400) from the STIC-Sophisticated Test & Instrumentation Centre at Cochin University of Science and Technology in Kerala was utilized. Confocal Raman Microscope with AFM (Alpha300RA AFM & RAMAN, Germany) at SAIF-Sophisticated Analytical Instrument Facility, School of Environmental Sciences, Mahatma Gandhi University, Kerala was used to record the Raman spectrum.

Ethical statement

With registration number 1174/ac/08/CPCSEA, the experiment was carried out strictly in accordance with the rules established by the Committee for Purpose of Control and Supervision of Experiments on Animals (CPCSEA). The Kerala University of Fisheries and Ocean Studies' institutional animal ethics committee thoroughly examined the protocol before approving it.

Experimental design

From the Kerala University of Fisheries and Ocean Studies hatchery, healthy juvenile *Etroplus suratensis* were acquired. Two weeks before the experiment began, the fish were acclimated to the lab environment (Dissolved oxygen = 4 to 5 ppm, Temperature = 27.83 ± 0.78 °C, pH = 7.51 ± 0.42, Alkalinity = 67 to 73 mg L^{-1}, Hardness = 64 to 69 mg L^{-1}). They were given artificial feed twice a day during acclimatization. Following acclimation, the fish, weighing an average of 9.46 ± 2.27 g, were split up into four groups for the MP accumulation experiment, each of which was replicated four times. Three fish in five liters of water were housed in each ten-liter aquarium (total no. of fish = 48). We used fluorescent PS-MPs for the MP bioaccumulation experiment. Fishes from four groups were subjected to exposure to fluorescent PS-MPs at concentrations of 0 (control), 0.2, 2, and 4 mg/L (treatment concentrations), respectively, over a period of 14 days. Similarly, the acclimated fishes were distributed into another four groups in four replicates for the MP toxicity experiment. Nine fishes in fifteen liters of water were housed in each twenty-liter aquarium. The fish were exposed to virgin PS-MPs at 0, 0.2, 2, and 4 mg/L for a duration of 14 days in the MP toxicity experiment. Both experiments received continuous aeration to ensure thorough dispersion of MP particles in the water. Throughout the 14 days semi-static exposure period, the exposure medium was refreshed every 2 days to sustain MP concentrations and eliminate waste buildup.

Samples collection

After a 14 days exposure period, a sample of eight randomly chosen fishes from each of the four groups was euthanized with MS 222 (Sigma-Aldrich, USA). The caudal severance method was used to obtain blood samples from the groups exposed to virgin PS MPs, which were then transferred to microcentrifuge tubes for biochemical analysis. The serum for biochemical analysis were separated and stored as mentioned by Das et al. 2023. Tissues from the intestine, liver, spleen, muscles, brain, and gills were taken from both experiments and preserved for MP bioaccumulation investigation at −80 °C in a sealed environment. 10% neutral buffered formalin (NBF) was used to fix tissues intended for confocal imaging. The liver tissue samples were processed and stored according to Das et al. 2023 to estimate oxidative stress in groups exposed to virgin PS-MPs. Liver tissue samples intended for gene expression analysis were aseptically collected from groups exposed to virgin PS-MPs and instantly stored in RNA later (Sigma-Aldrich, USA) at −80 °C for subsequent analysis.

Analysis of the MP bioaccumulation

Three techniques, including fluorescence measurements, microscopic enumeration, and confocal microscopy, were employed to assess MP bioaccumulation in various tissues. Detection and quantification of fluorescent MPs in the intestine, gills, liver, spleen, muscle, and brain were conducted by using confocal microscope and multimode plate reader (Tecan Infinite® 200 PRO) respectively. Confocal images of the ingested MPs in different tissues of pearl spot were taken with a laser scanning confocal microscope (Nikon A1-HD) at RGCB (Rajiv Gandhi Centre for Biotechnology, Thiruvananthapuram, Kerala) (excitation wavelength: 488, emission wavelength: 525 nm). Before microscopic observation, the tissue samples were subjected to thin sectioning and placed in a 35 mm confocal grade glass bottom dish. For the determination of MP concentrations in tissues, KOH-based tissue digestion was performed following the methods of (Li et al. 2018; Ding et al. 2020), and (Huang et al. 2021). 50 mg of tissue samples were digested in 5 mL KOH solution (100 g/L) at 60 °C for 72 h. Deionized water was used to dilute the digestions until a final volume of 10 mL was achieved. The multimode plate reader (excitation and emission at 470 and 505 nm, respectively) was used to directly measure the fluorescence intensity of the digested solutions. In compliance with the standard curve, the fluorescent MPs were quantified. The background fluorescence of various tissues of control fish was subtracted from that of exposed samples.

Further, the bioaccumulation of virgin MPs in these tissues was determined by counting the MP particles in the tissue digestions under the light microscope. The tissues of virgin MPs exposed groups were also subjected to KOH digestion as described above. A light microscope was then used to count 100 μl of the digested solution that had been fixed in a sanitized glass slide.

Analysis of biochemical parameters

Agappe Diagnostic kits (Agappe Diagnostics Ltd, Kerala, India) were used to estimate serum biochemical parameters such as glucose, total protein, A/G ratio, total cholesterol, SGOT, SGPT, and ALP. Using a multimode plate reader, the absorbance was measured at related wavelengths after the parameters were established in accordance with the manufacturer's instructions.

Measurement of cortisol

In accordance with the manufacturer's instructions, commercial ELISA kits acquired from Origin Diagnostics and Research, India, were used to measure the serum cortisol level.

Oxidative stress analysis

In liver tissue homogenate, oxidative stress markers such as SOD, CAT, GPx, TAC, PC, and MDA were measured.

SOD was ascertained using the technique elucidated by (Mishra and Fridovich 1972). Estimates of catalase were made using the method illustrated by (Takahara et al. 1960). The Bradford method (Bradford 1976) was used to estimate the protein content of liver tissues. TAC was measured by the Cupric ion reducing antioxidant capacity (CUPRAC)

Table 1 Primers used for qPCR analysis

Gene name	Primer sequence (5′→3′)	Primer Efficiency (%)	References
18S rRNA	F: GGACACGGAAAGGATTGACAG R: GTTCGTTATCGGAATTAACCAGAC	98.7	(Wang et al. 2015)
EF1α	F: CTGTACCCGTCGGTCGTGTT R: GGTGCATCTCCACGGACTTC	96.7	(Bhat et al. 2015)
β-ACTIN	F: CCTGAGCGTAAATACTCCGTCTG R: AAGCACTTGCGGTGGACGAT	95.9	(Liang et al. 2018)
HSP70	F: CTGTGGAGAAGTCGCTCCGT R: GACAGCAGCTCCATAGGCCA	95.3	This study
IGF1	F: GACGAATGCTGCTTCCAAAG R: GCCCTTGTCCGCTCTACT	97.2	This study
CYP1A	F: ACCGCTGCATCGTTGTCTCT R: ATCGTGTTGCTGGGCAGGTA	98.3	This study
NRF2	F: CTGCCGTAAACGCAAGATGG R: ATCCGTTGACTGCTGAAGGG	96.3	(Zheng et al. 2021)
P53	F: GCATGTGGCTGATGTTGTTC R: GCAGGATGGTGGTCATCTCT	96.4	(Farag et al. 2021)

method described by (Ardhani et al. 2019). The GPx, MDA activity and PC content were determined using the commercial assay kits acquired from Origin Diagnostics and Research, India based on the instructions of the kit manufacturer.

Analysis of gene expression

TRI Reagent (Sigma Aldrich, USA) was used to extract total RNA from the liver, and a Nano-Drop ND-100 UV–Vis spectrophotometer (NanoDrop Technologies LLC, Wilmington, USA) was used to measure the amount of RNA. RNA was then exposed to DNase I (amplification grade, 1 unit/μg RNA, Invitrogen). Using the cDNA synthesis kit with RNase inhibitor (Origin Diagnostics and Research, India), reverse transcription was carried out from 2 μg of total RNA in accordance with the manufacturer's instructions.

The online tool: Primer-BLAST was used to design the primers for *HSP70*, *IGF1*, and *CYP1A* (Ye et al. 2012), except the primers for *NRF2*, *P53*, and reference genes which were obtained from the literature. Each primer pair's PCR amplification efficiency was calculated using the standard curve slopes. The expression data of the three reference genes *18S rRNA*, *EF1α* and *β-ACTIN* were first analyzed by four currently used software tools and then by RefFinder to determine the best reference gene.

In the CFX96 Real-time PCR Detection System (Bio-Rad Laboratories, USA), quantitative real-time PCR was carried out using the SYBR green PCR master mix (Origin Diagnostics and Research, India). The primer sequences for the internal standard genes and target genes are displayed in Table 1. For the reaction, the thermocycling conditions were 30 s at 95 °C, followed by 40 cycles of 5 s at 95 °C and 1 min at 60 °C. The annealing temperature was 63.5 °C for *IGF1*. Using the $2^{-\Delta\Delta Ct}$ method, the relative fold changes in gene expression were examined (Livak and Schmittgen 2001).

Statistical analysis

Before analysis, all of the data were examined using Shapiro-Wilk'stest and Levene's test to ensure that the variables were normal and that the variance was homogeneous. One-way ANOVA was used to analyze the data using SPSS 22.0 (SPSS version 22, SPSS Inc., Il, USA).

To ascertain treatment differences, Duncan's multiple-range test was employed, with a significance level of $P < 0.05$. Every value is given as mean ± standard error of mean (SEM).

Results

Characterization of the microplastic

Figure 1 illustrates the spherical shape of the MP with an average diameter of 1 ± 0.111 μm, as confirmed by the SEM image. The Raman spectrum (Fig. 2) confirmed the composition of the MP as polystyrene.

Detection of PS-MPs in various tissues

The quantification of MP bioaccumulation in terms of weight and count were carried out using fluorescent and virgin MPs respectively. Pearl spot exposed to 0.2, 2 and 4 mg/L of PS-MPs showed MP bioaccumulation in their intestine, gills, liver, spleen, muscle, and brain. It was discovered that the order of PS-MP bioaccumulation in pearl spot tissues was intestine, gills, liver, spleen, muscle, and

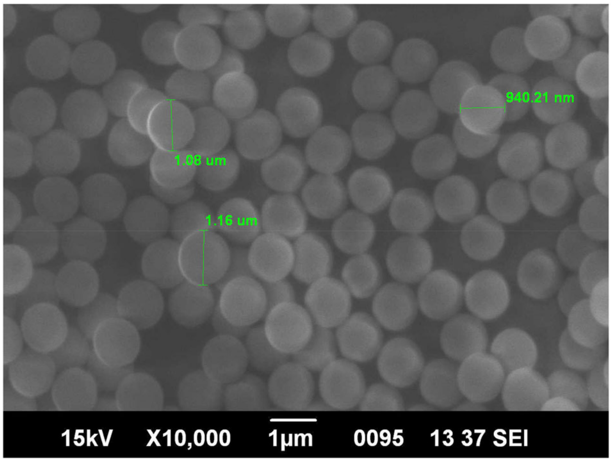

Fig. 1 A scanning electron microscope (SEM) image (×10,000) of PS-MP

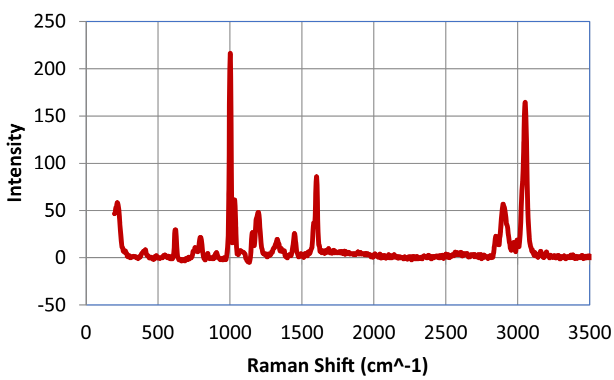

Fig. 2 Raman spectrum for PS-MP

brain (Figs. 3, 4, 5 and 6). The study clearly demonstrated that at the highest MP exposure concentration of 4 mg/L, the bioaccumulation of MP in the gills is 3.9 times lower than in the intestine. Furthermore, at the same exposure concentration, bioaccumulation is significantly lower in other organs: 9.6 times lower in the liver, 16.9 times lower in the spleen, 70.6 times lower in the muscle, and an astonishing 155.6 times lower in the brain compared to the intestine.

Serum biochemical parameters

After being treated with 2 and 4 mg/L of PS-MPs for 14 days, there was a significant increment in the serum biochemical parameters, including glucose, total cholesterol, SGPT, and ALP ($p < 0.05$). All the MP treatment groups showed significantly high serum levels of SGOT and total protein ($p < 0.05$). There was a noteworthy reduction in both albumin and A/G ratio in the groups treated with MP, as compared to the control group ($p < 0.05$). However, there was no significant difference found between the groups treated with 2 and 4 mg/L of MP (as shown in Fig. 7).

Effect on cortisol level

After 14 days of exposure, the MP exposure significantly raised cortisol levels in response to the MP concentration ($p < 0.05$) (Fig. 8).

Oxidative stress parameters

When compared to the control, a significant inhibition in the SOD, CAT, and GPx activities as well as a reduction in total antioxidant capacity were noticed in the MP-exposed pearl spot livers ($p < 0.05$). Nevertheless, no discernible difference ($p > 0.05$) in the decline of hepatic CAT activity and total antioxidant capacity were observed between the 0.2 mg/L MP exposure group and the control group (Fig. 9). After 14 days, all three treatment groups had significantly higher liver MDA and protein carbonyl contents ($p < 0.05$) due to the PS-MP (Fig. 9).

Gene expression

After evaluating three reference genes (*18S rRNA*, *EF1α*, and *β-ACTIN*), *18S rRNA* was found to be the most stable and was used as the reference gene for expression analysis. In all the MP-exposed groups, in a concentration-dependent way, hepatic mRNA expression of *HSP70* was significantly up-regulated ($p < 0.05$) when compared to the control. As the concentration of PS-MP increased, it was noted that the expression of *IGF1*, *CYP1A*, *NRF2*, and *P53* was significantly down-regulated in the liver ($p < 0.05$) (Fig. 10).

Discussion

According to our findings, juvenile pearl spot fish that were exposed to 1 µm PS-MPs at concentrations that are commonly found in the environment for fourteen days exhibited the highest level of microplastic bioaccumulation in their intestines, followed by their gills, liver, spleen, muscle, and brain, respectively. After being treated with 300 mg/L PS-MPs of size 8 µm for 7 days, MP accumulation was observed in different tissues of goldfish following the order of gills > intestine > liver (Abarghouei et al. 2021). However, in adult zebrafish, 5 µm PS-MPs exhibited the highest accumulation in the gastrointestinal tract (GIT) which is followed by the gill. In contrast, no MPs were detected in the brain and muscle after 7 days exposure (Chen et al. 2020). Similarly, at 100 µg/L after 14 days, the gut of red tilapia had the highest accumulation of PS-MPs of 5 µm size. Further, the magnitude of PS-MP accumulation in red

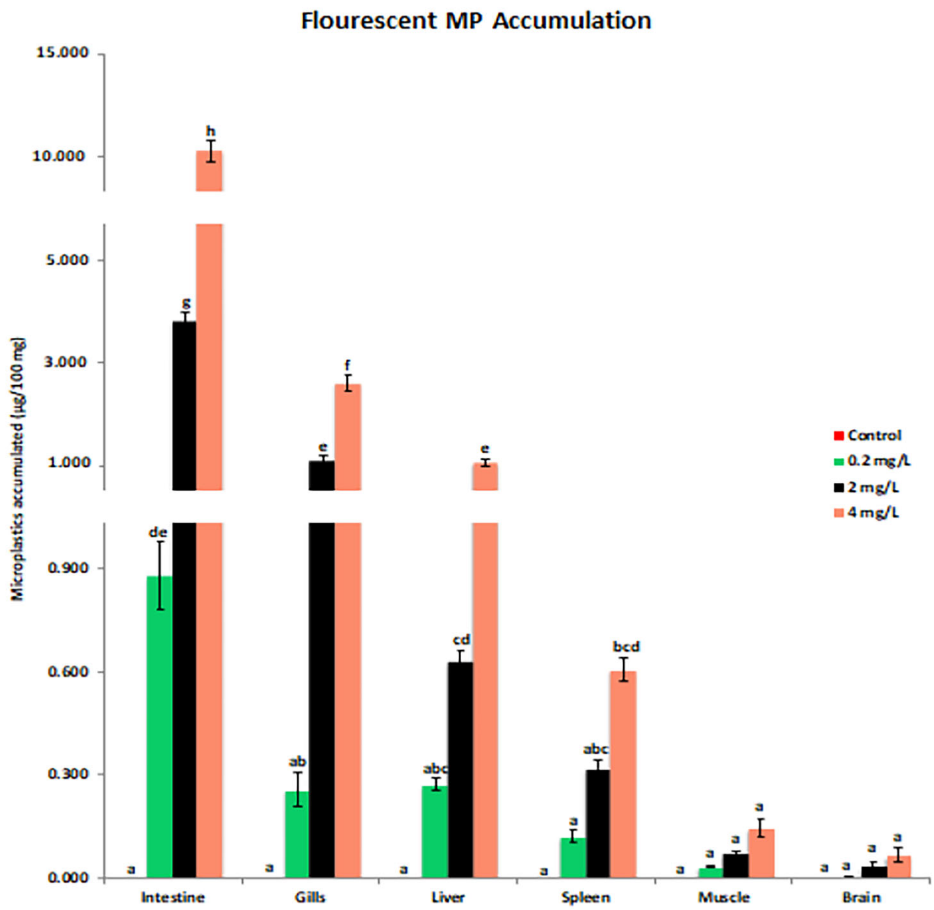

Fig. 3 Bioaccumulation of flourescent PS-MP particles (µg/100 mg) in *E. suratensis* tissues after 14 days exposure ($p < 0.05$)

tilapia tissues showed a variation in such a way that after 6 days exposure the order of accumulation was gills > brain > liver > gut and that after 14 days exposure followed the order of gut > liver > brain > gills (Ding et al. 2020). MPs at the size of ≤20 µm were found in the gastrointestinal tract, gills and head of juvenile Japanese medaka after 14 days exposure. In addition, it has been revealed that in Japanese medaka the bioconcentration factor (BCF) for smaller MPs of size 2 µm is higher compared to larger MPs (20 and 200 µm MPs) (Liu et al. 2021). Small MPs, measuring less than 3 µm, were found in muscle and gastrointestinal tract of adult benthopelagic fish, *Serranus scriba*, that were captured off the coasts of Tunisia (Zitouni et al. 2020). Early juvenile European sea bass were exposed to smaller environmental MPs (0.45–3 µm) for 3 and 5 days through their diet. As a result, MP was accumulated in the gills, gut, and liver, and liver exhibited the highest quantity (Zitouni et al. 2021). Following a 7-day exposure to 20 mg/L, PS-MPs of 5 µm size were detected in the gills, liver, and gut of zebrafish, whereas 20 µm PS-MPs were found exclusively in the gills and gut (Lu et al. 2016). MPs were also reported from the muscle of wild European seabass, Atlantic horse mackerel, and Atlantic chub mackerel from Portuguese coastal waters. Moreover, the human intake of MPs through fish consumption was assessed to be between 112 and 842 MP items/g/year (Barboza et al. 2020). After two months of chronic exposure to PS-MPs (10 µm) at environmentally relevant concentrations, MPs were accumulated in gills, intestines, and livers of marine medaka (Wang et al. 2019). Estuarine and marine fish unequivocally consume more water than freshwater fish to effectively maintain their osmotic balance in higher ionic environments (Greenwell et al. 2003; Marshall. 2012; Griffith. 2017). This heightened water intake is a critical factor in microplastic toxicity, as it significantly increases the ingestion of microplastics and exposure to associated chemicals. As a result, the toxic effects of microplastics in estuarine and marine fish may be considerably greater compared to those in freshwater fish. The 1 µm sized PS-MPs were observed to accumulate in several tissues after 14 days in our previous study on Nile tilapia (*Oreochromis niloticus*). MP bioaccumulation was observed in different tissues of Nile tilapia following the order of intestine > gills > liver > spleen > muscle ≈ gonad > brain (Das et al. 2023). In other words, the brain and intestine had the lowest and maximum bioaccumulation, respectively, which is consistent with the current study's

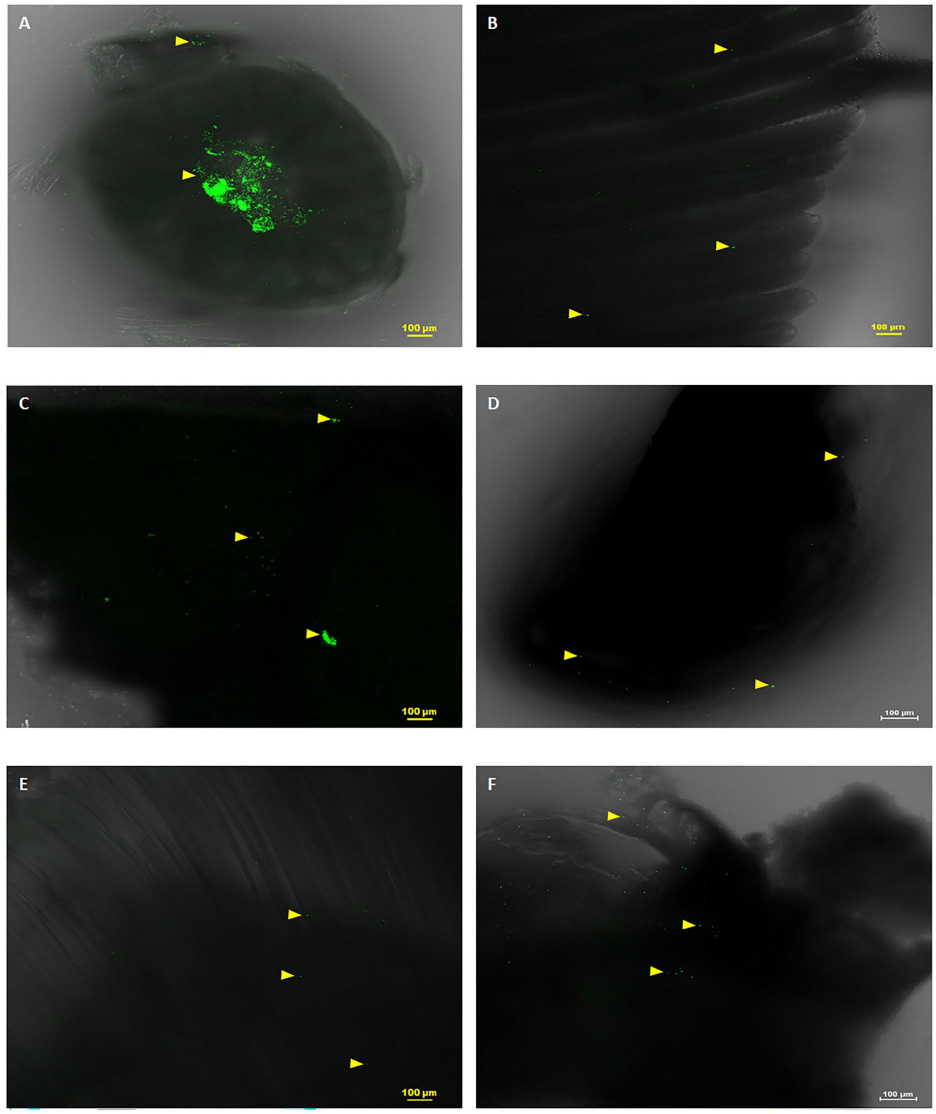

Fig. 4 Confocal microscopic images of fluorescent PS MPs bioaccumulated in different tissues of *Etroplus suratensis* after 14 days of exposure. **A** Intestine, **(B)** Gills, **(C)**Liver, **(D)** Spleen, **(E)** Muscle and **F** Brain

findings. The juvenile loach's liver accumulated more MPs than the gill and intestine after being exposed to 1000 g/L of 5 m PS-MPs for 7 days. After being exposed to PS-MPs (5 µm size, at 1000 µg/L) for one-week, juvenile loach showed a higher accumulation of MPs in the liver compared to the gill and intestine (Wang et al. 2022).

The results of the present study are consistent with previous research, which indicates that MP bioaccumulation in fish primarily occurs in the intestine, gills, and liver. The intestine and gills serve as the main entry points for MPs into fish, while the liver acts as the primary organ responsible for filtering and metabolizing environmental pollutants. As a result, these organs tend to bioaccumulate higher levels of MPs. The feeding habits of fish have been shown to influence the bioaccumulation of MPs and NPs in their tissues (Chen et al. 2025). As an omnivorous fish, *Etroplus suratensis* has a longer intestine, which in turn, raises the likelihood that microplastics will be bioaccumulated more in the intestine. Furthermore, this study found that the muscle of *Etroplus suratensis* is bioaccumulated with MPs sized at 1 µm. This indicates the potential for smaller MPs to translocate into fish muscle. Consequently, this suggests a possible risk of these smaller MPs being bioaccumulated in other organisms and humans at higher trophic levels through the food chain.

According to the current study, glucose, cholesterol, and total protein were remarkably elevated in pearl spot, while albumin and A/G ratio decreased in the PS-MPs exposed group. During times of stress, carbohydrates serve as an instant energy source, causing blood glucose levels to increase as a result of glycogenolysis (Dobšíková et al. 2009; Javed and Usmani 2015). However, 1 mg/L small-sized PS-MPs caused a decrease in muscle glucose content in adult zebrafish after 7 days. The low glucose levels may

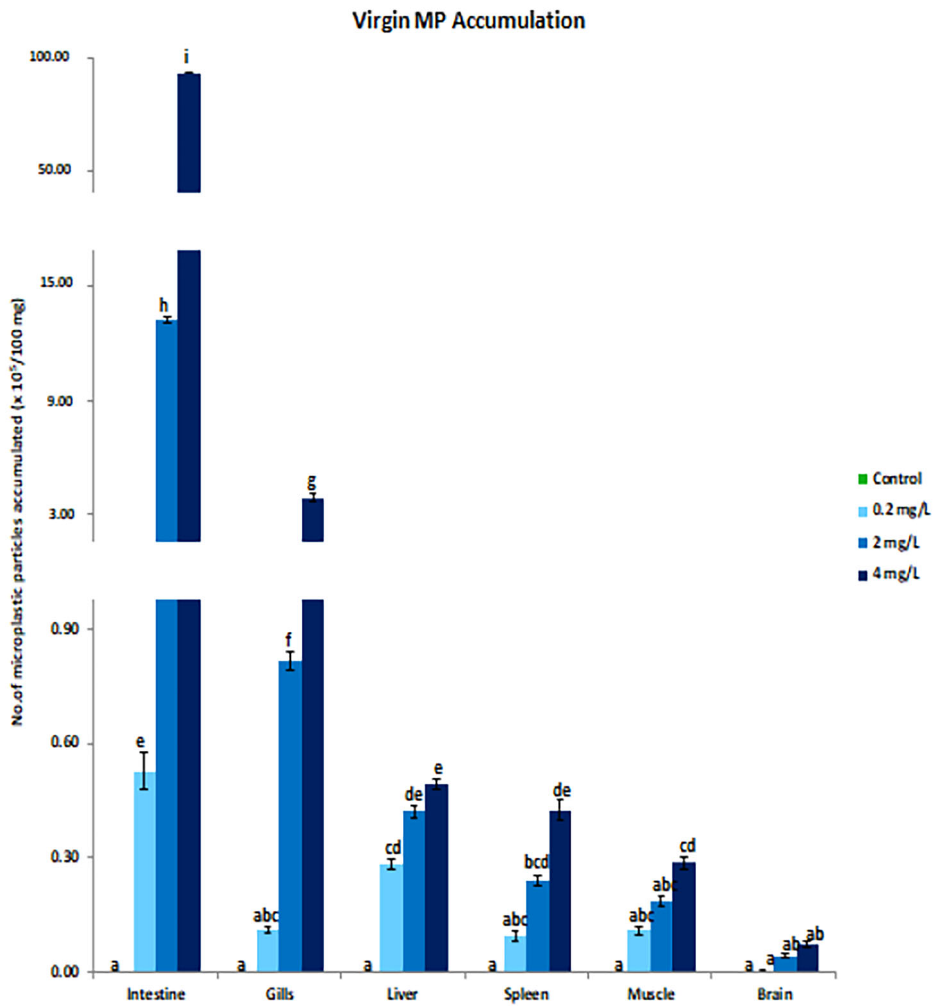

Fig. 5 Bioaccumulation of virgin PS-MP particles ($\times 10^5$/100 mg) in *E. suratensis* tissues after 14 days exposure ($p < 0.05$). Different letters (**a**, **b**, **c**, **d**, **e**, **f**, **g**, **h**, **i**) denote statistically significant differences between groups.

be attributed to energy depletion in fish after MP exposure (Chen et al. 2020). Most of the proteins in the fish body are albumin and globulin. When intoxicated with xenobiotics, changes in the quantities of these proteins can result in a disturbed A:G ratio (Javed and Usmani 2015). In juvenile common carp the plasma content of total protein, globulin, and albumin levels were depleted after one month chronic MPs exposure (250 and 500 µg/L). However, the levels of glucose and cholesterol were increased. Elevated cholesterol occurs when lipid is mobilized for stress relief after glycogen and carbohydrate depletion (Javed and Usmani 2015; Lind and Lind 2020). Increased AST, ALT, and ALP levels could be a sign of liver tissue damage (Kandemir et al. 2018; Hamza et al. 2023). During our study, hepatic damage caused by the MP exposure likely resulted in an elevation of AST, ALT, and ALP levels in pearl spot. Similarly, the 30 days of MP exposure (250 and 500 µg/L) increased AST, ALT and ALP activities in juvenile common carp (Banaee et al. 2019). The activity of ALP showed a significant increase in common carp treated with 2 mg/L MPs for 21 days. However, exposure to MPs for 21 days at 1 and 2 mg/L resulted in a significant decline in the levels of total protein, albumin, globulin and cholesterol with no significant variations in glucose, AST and ALT contents in common carp (Nematdoost and Banaee 2017). Glucose, cholesterol, albumin, globulin, A/G ratio, ALT, AST, and ALP displayed noteworthy ($p < 0.05$) increment in Nile tilapia early juvenile after 15 days of MP treatment at 1 mg/L (Hamed et al. 2019). According to (Das et al. 2023), for 14 days, Nile tilapia was subjected to PS-MPs (1 µm) exposure at environmentally relevant concentrations which increased total protein, glucose, A/G ratio, ALP, SGPT, and SGOT. In summary, exposure to smaller MPs at environmentally relevant concentrations can induce changes in biochemical parameters due to stress and liver tissue damage in fish.

Because antioxidants facilitate cell detoxification, they are essential for shielding cells from oxidative stress (Bainy 1996; Biller and Takahashi 2018). Our findings indicate that exposure to MPs inhibited antioxidant enzymes (SOD,

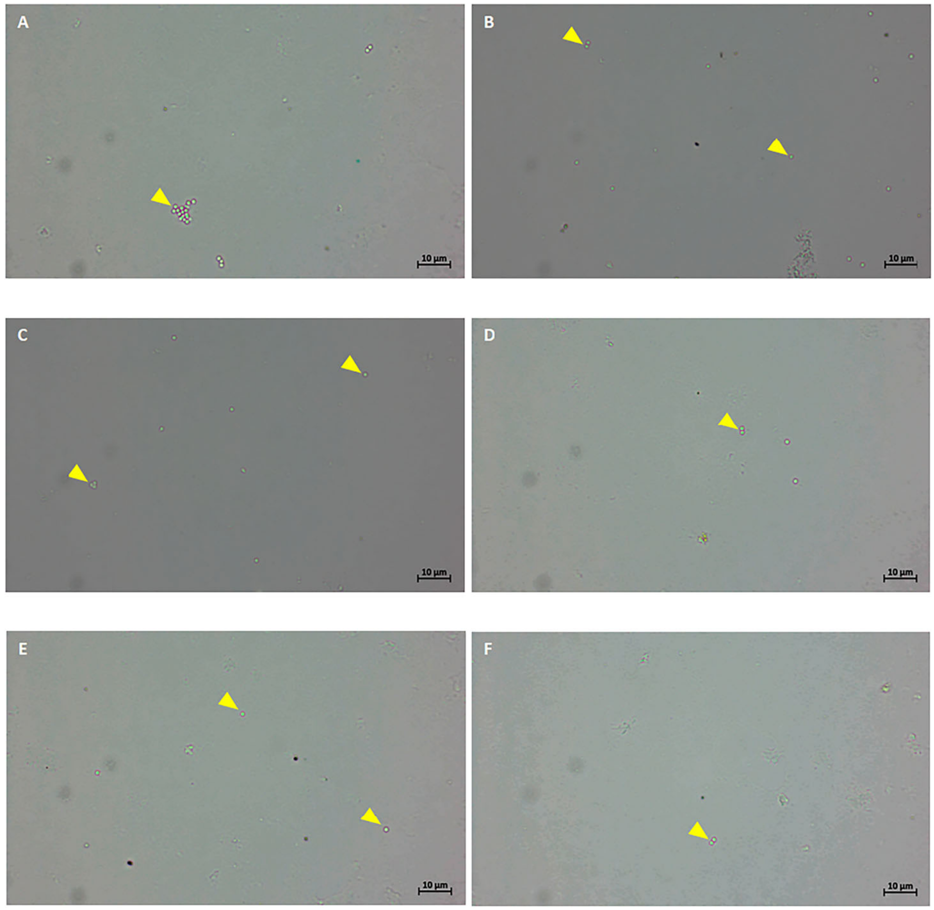

Fig. 6 Microscopic images (×100) of virgin PS MPs bioaccumulated in different tissues of *Etroplus suratensis* after 14 days of exposure. **A** Intestine, (**B**) Gills, (**C**) Liver, (**D**) Spleen, (**E**) Muscle and **F** Brain

CAT, and GPx), diminished TAC levels, and induced oxidative damage by increasing MDA and protein carbonyl levels. Similar to our results, gold fish exposed to 5 mg/L polystyrene MPs with size 8 µm for 28 days suppressed the hepatic expression of SOD and CAT genes. However, serum's SOD and CAT activity was found to be elevated by MP exposure (Abarghouei et al. 2021). Adult zebrafish did not show any significant changes in their hepatic CAT, SOD, and GPx activities due to exposure to 1 mg/L of PS-MPs (5 µm) (Chen et al. 2020). In red tilapia, increased SOD activity and MDA content in the liver have been recorded after 14 days of treatment with 5 µm PS-MPs at a concentration of 100 µg/L (Ding et al. 2020). An elevated ROS level indicates oxidative stress induced by MPs in adult zebrafish gonads after exposure to PS-MPs (1 µm; 100 and 1000 µg/L) for 21 days (Qiang and Cheng 2021). Even 48 h exposure to low MP (5 µm PS-MP) concentration of 80 µg/L induced oxidative stress in juvenile silver carp by elevating SOD activity and MDA level (Zhang et al. 2021a). Oxidative stress was evident in early juvenile European sea bass after 3 and 5 days dietary exposure to small environmental MPs (0.45–3 µm) through the elevation of CAT, SOD and GST activities in the gut, liver, and gills (Zitouni et al. 2021). Short-term (96 hour) exposure to smaller MPs at 0.69 mg/L caused oxidative damage in juvenile European seabass (Barboza et al. 2018). After 30 days, the SOD, CAT, GPx activities and MDA levels were increased in juvenile discus fish subjected to PS-MPs (50 and 500 µg/L). However, the 30 days MPs exposure didn't cause any variations in protein carbonyl content (Wen et al. 2018). Lipid oxidative damage was noted in gills and muscle of wild fish from Portuguese coastal waters in relation to MP contamination (Barboza et al. 2020). SOD, CAT, total peroxides, and oxidative stress index (OSI) went up in Nile tilapia early juveniles after 15 days of exposure to MPs (1 mg/L). In contrast, TAC decreased in groups subjected to 1 mg/L of MPs (Hamed et al. 2020). PS-MPs (1 µm) exposure (at environmentally relevant concentrations) were noted to reduce the SOD activity and induce the CAT activity and MDA levels of Nile tilapia after 14 days (Das et al. 2023). Following a 21-day exposure period to 5 µm PS-MPs (at exposure concentrations100 and 1000 µg/L), the activity of SOD, CAT, and GPx was notably reduced in juvenile loach (Wang et al. 2022). Our

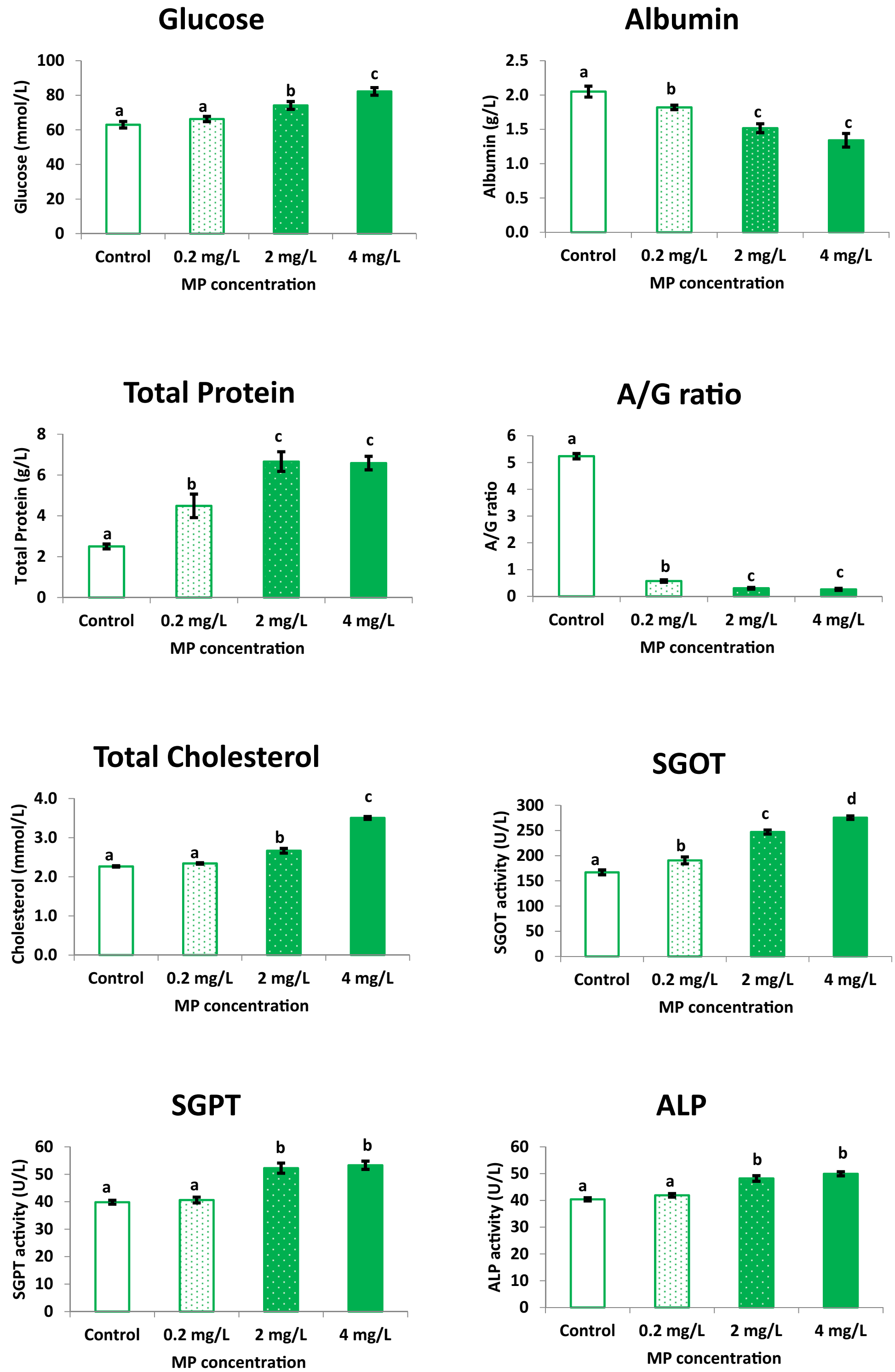

Fig. 7 Serum biochemical parameters such as glucose, albumin, total protein, A/G ratio, total cholesterol, SGOT, SGPT and ALP of *E. suratensis* exposed to 0, 0.2, 2 and 4 mg/L of PS-MPs concentrations over a period of 14 days ($p < 0.05$). Different letters (**a**, **b**, **c**, **d**) denote statistically significant differences between groups.

results indicate that exposure to smaller MPs at environmentally relevant concentrations weakens the antioxidant system, making fish more susceptible to oxidative damage.

In this study, fish exposed to MP depicted a severe decrease in *NRF2* and *P53* expression. This reduction reflects the failure of ROS detoxification. Similarly, after 2 weeks of 5 μm PS-MPs treatment at concentrations of 100 and 1000 μg/L, significant down-regulation of *NRF2* was observed in juvenile loaches (Wang et al. 2022). According to (Wang et al. 2022), long-term exposure to MP may cause liver damage, resulting in *NRF2* down-regulation. No changes in *TP53* gene expression were observed in

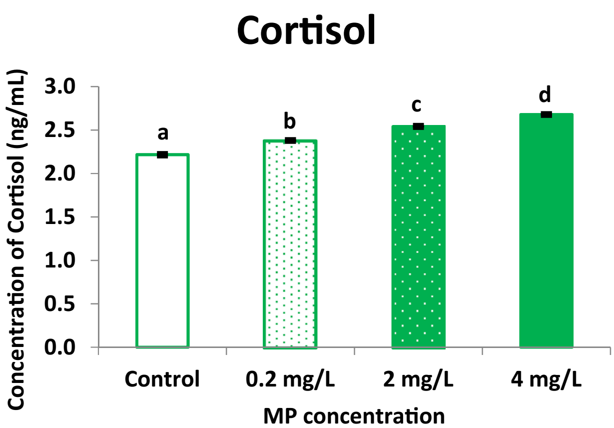

Fig. 8 Effect on cortisol level of *E. suratensis* exposed to 0, 0.2, 2 and 4 mg/L of PS-MPs concentrations over a period of 14 days ($p < 0.05$). Different letters (**a**, **b**, **c**, **d**) denote statistically significant differences between groups.

zebrafish larvae following exposure to virgin LDPE fragments for 10 and 20 days (Karami et al. 2017). Conversely, the testis of adult zebrafish and the liver of Nile tilapia exposed to PS-MPs (1 µm size; at 0.1 and 1 mg/L concentrations) showed a notable upregulation of *P53* (Qiang and Cheng 2021; Das et al. 2023). Previous studies have identified that MP induced oxidative damage in fish is associated with ROS-mediated p53 signaling pathway, Keap1-Nrf2 signaling pathway and oxidative stress mediated endoplasmic reticulum pathway (Umamaheswari et al. 2021; Wang et al. 2022; Xue et al. 2022; Yao et al. 2023).

In this study, we have found that MP toxicity evokes oxidative stress, leading to a substantial diminish in antioxidant indices such as SOD, CAT, GPx, and TAC. Additionally, we observed reduced expression of *NRF2* and *P53* genes while there was a significant augmentation in MDA and protein carbonyl contents. The current study's

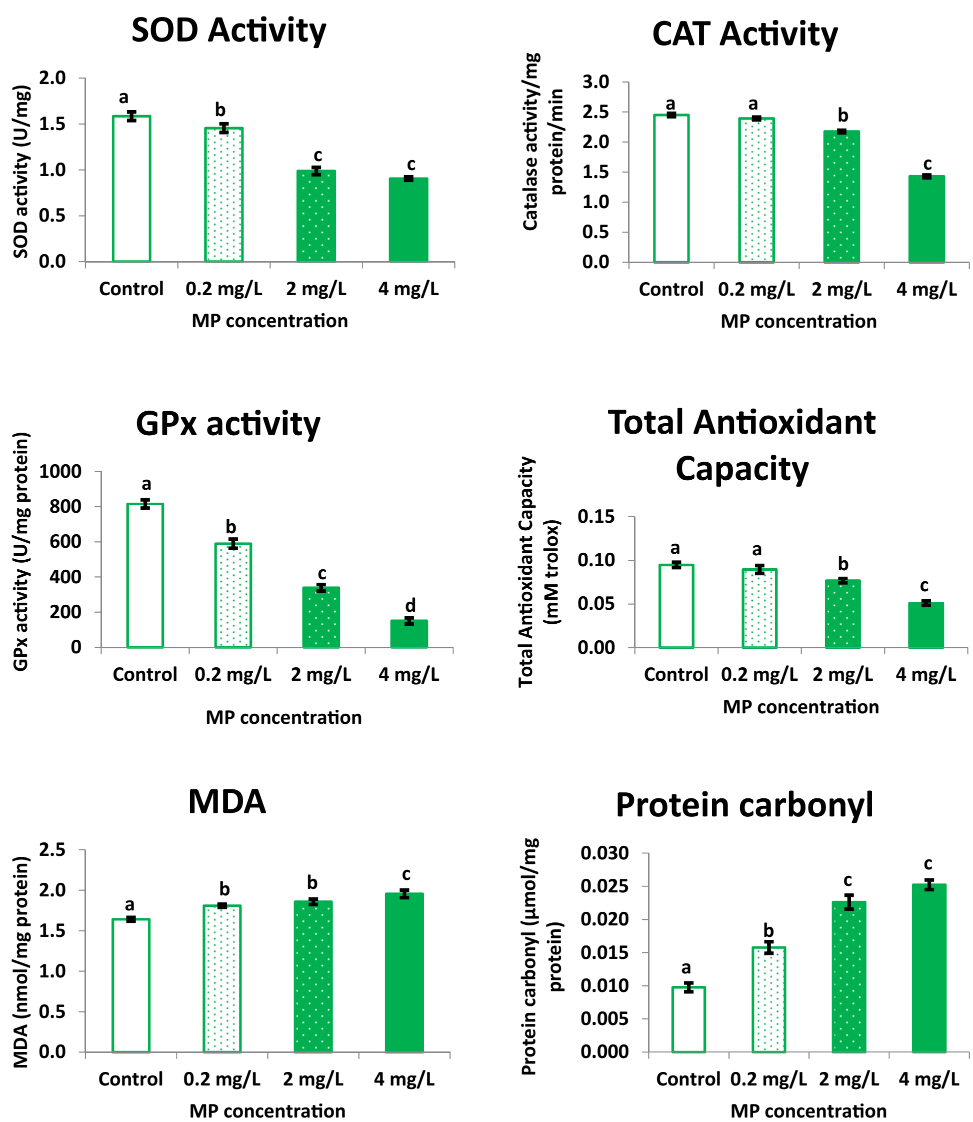

Fig. 9 Oxidative stress parameters such as SOD, CAT, GPx, total antioxidant capacity, MDA and protein carbonyl of *E. suratensis* exposed to 0, 0.2, 2 and 4 mg/L of PS-MPs concentrations over a period of 14 days ($p < 0.05$). Different letters (**a**, **b**, **c**, **d**) denote statistically significant differences between groups.

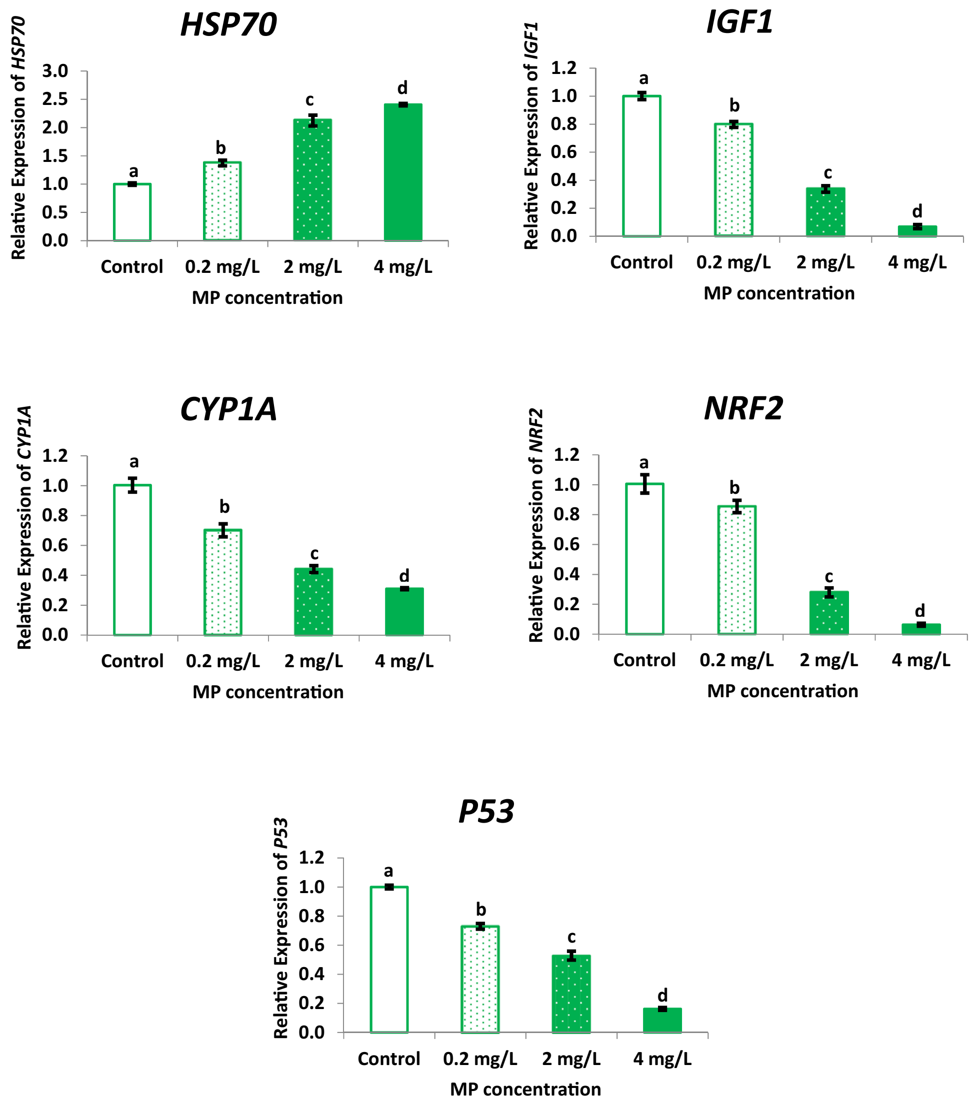

Fig. 10 Effects on the expressions of *HSP70*, *IGF1*, *CYP1A*, *NRF2* and *P53* in the liver of *E. suratensis* exposed to 0, 0.2, 2 and 4 mg/L of PS-MPs concentrations over a period of 14 days ($p < 0.05$). Different letters (**a**, **b**, **c**, **d**) denote statistically significant differences between groups.

findings show that fish ROS detoxification systems can be harmed by MP exposure at environmentally relevant concentrations.

In the current study, we reveal the occurrence of a stress condition elicited by MP exposure through the elevated levels of stress indicators such as glucose, cortisol, and the gene expression of *HSP70*. Several studies have revealed that MP toxicity can lead to increased levels of cortisol and glucose, as well as an up-regulated *HSP70* expression. These findings were supported by (Banaee et al. 2019; Hamed et al. 2019; Abarghouei et al. 2021; Zhang et al. 2021a), and (Das et al. 2023). Consistent with the findings of this investigation, at environmentally relevant concentrations, the Nile tilapia showed a significant increase in cortisol levels following a 14-day exposure to PS-MPs measuring 1 µm (Das et al. 2023). In marine medaka, exposure to MP caused a cortisol-mediated stress response by disrupting the regulation of the hypothalamus-pituitary-interrenal (HPI) axis (Zhang et al. 2024). Goldfish exposed to PS-MPs (8 µm) for 28 days at 0.05 and 0.5 mg/L were shown to have an upregulated hepatic *HSP70* gene (Abarghouei et al. 2021). Up-regulated *HSP70* gene expression was observed in the intestine of juvenile silver carp after 48 h exposure to 80 µg/L PS-MPs (5 µm) (Zhang et al. 2021a). Our earlier research on Nile tilapia also demonstrated that exposure to MPs at environmentally relevant concentrations for 14 days could upregulate the expression of the *HSP70* gene (Das et al. 2023). These results indicate that *HSP70* genes in fish are highly sensitive to MP toxicity. Overall, the research on *HSP70* reveals its role in mitigating MP-induced stress in fish. However, the mechanism underlying MP-induced regulation of *HSP70* in fish, has to be further studied.

According to (de Souza Anselmo et al. 2018), cytochrome P450 (CYP)-dependent enzymes are essential for the oxidative metabolism of a range of xenobiotics during detoxification. According to the current study, pearl spot's *CYP1A* gene expression was downregulated as a result of MP exposure. This suggests that a higher accumulation of MPs in the liver could have disrupted the fish's hepatic detoxification metabolism. However, 5 µm PS-MPs (100 µg/L) exposure for 14 days elevated the CYP1A-associated enzyme EROD activity in red tilapia (Ding et al. 2020). Acute exposure to high-density polyethylene MPs caused significant up-regulation of *CYP1A* in the adult zebrafish intestine (Mak et al. 2019). Further, MP-induced *CYP1A* up-regulation was observed in larval stickleback exposed to PS-MPs (1 µm) for 7 days (Katzenberger and Thorpe 2015). Similar to the results of our study, chronic PS-MPs (2 µm) exposure was found to disrupt the metabolism of xenobiotics by depressing cytochrome P450 pathways in marine medaka (Wang et al. 2021). Furthermore, after sub-chronic exposure to PS-MP at environmentally relevant concentrations, the expression of the *CYP1A* gene was significantly down-regulated in the liver of Nile tilapia (Das et al. 2023). The metabolism of xenobiotics by cytochrome P450 is a significant pathway activated in zebrafish larvae upon exposure to MPs (Wang et al. 2024). According to this study, higher microplastic bioaccumulation can negatively affect the metabolism pathways of xenobiotics in fish.

IGF1, a member of the IGF family, functions to regulate growth (Duan 1998). Our research revealed that, in comparison to the control group, MP exposure dramatically reduced the expression of the *IGF1* gene in pearl spot liver. Furthermore, in marine medaka larvae, long-term exposure to PS-MPs (2 µm) (parental exposure to 20 and 200 µg/L MP concentration) reduced body length and down-regulated the expression of *GHR*, *IGF-I*, and *IGF-IR* (Wang et al. 2021). On the other hand, after 60 days MP treatment at 10 mg/L, the effects of 2 and 10 µm PS-MPs on marine medaka's body weight and body length were not statistically significant (Zhang et al. 2021b).

Altogether, the higher bioaccumulation of smaller MPs observed in *E. suratensis* led to a distinct array of toxic impacts. This includes liver tissue damage accompanied by changes in biochemical parameters, a reduced antioxidant response leading to oxidative damage, and disruption of toxin metabolism. These MP-induced effects in brackish water fish species require further investigation to determine if they are ecologically specific.

Conclusion

This study represents an in-depth investigation into the buildup of ecologically significant levels of smaller microplastics in six distinct tissues of a brackish water fish species. Additionally, it explores the impact of this bioaccumulation on various antioxidant indicators following sub-chronic exposure to smaller microplastics through water. To the best of our knowledge, no previous research has looked at the toxicity and bioaccumulation of environmentally relevant amounts of smaller PS-MPs in pearl spot. Our investigation aimed to fill this gap by quantifying the bioaccumulation of 1 µm sized smaller MPs in various *Etroplus suratensis* tissues. This is the first detailed study conducted to assess the MP bioaccumulation in fish using three techniques such as fluorescence measurements using fluorescence-labelled PS-MP, confocal microscopy, and microscopic enumeration. Our findings revealed a distinct pattern of Smaller PS-MPs bioaccumulation, with the highest level observed in the intestine, followed by the gills, liver, spleen, muscle, and brain. In each tissue, the MP bioaccumulation was quantified as count/100 mg tissue and in µg/100 mg tissue. In order to enhance our understanding of how microplastics affect fish antioxidant systems, we examined a range of variables, including SOD, CAT, GPx, TAC, PC, and MDA levels, as well as the expression patterns of the *NRF2* and *P53* genes. This study represents the primary endeavor to examine the entirety of these factors, offering a comprehensive insight into their interaction. Sub-chronic exposure to MPs resulted in significant alterations in serum biochemical parameters, induction of oxidative stress, activation of stress responses, and down-regulation of the expression of *IGF1*, *CYP1A*, *NRF2*, and *P53* in pearl spot juveniles. Moreover, our study underscores the potential health risks associated with consuming fish muscle that has bioaccumulated smaller MPs. Additionally, our research sheds new light on the ecological threat posed by smaller MP pollution to the long-term sustainability of fresh/brackish water fish populations.

Data availability

No datasets were generated or analysed during the current study.

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Author contributions Bini C Das: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing - original draft, Writing - review & editing. Vineetha VP: Formal analysis, Methodology, Software. Devika Pillai: Funding acquisition, Resources, Project administration. Rejish Kumar VJ: Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Validation, Visualization, Writing - original draft, Writing - review & editing.

Compliance with ethical standards

Conflict of interest The authors declare no competing interests.

References

Abarghouei S, Hedayati A, Raeisi M, Hadavand BS, Rezaei H, Abed-Elmdoust A (2021) Size-dependent effects of microplastic on uptake, immune system, related gene expression and histopathology of goldfish (*Carassius auratus*). Chemosphere 276: 129977. https://doi.org/10.1016/j.chemosphere.2021.129977.

Abidli S, Pinheiro M, Lahbib Y, Neuparth T, Santos MM, Trigui El Menif N (2021) Effects of environmentally relevant levels of polyethylene microplastic on *Mytilus galloprovincialis* (Mollusca: Bivalvia): filtration rate and oxidative stress. Environ Sci Pollut Res Int 28:26643–26652. https://doi.org/10.1007/s11356-021-12506-8.

Anagha PL, Viji NV, Devika D, Ramasamy EV (2023) Distribution and abundance of microplastics in the water column of Vembanad Lake–a Ramsar site in Kerala, India. Mar Pollut Bull 194: 115433. https://doi.org/10.1016/j.marpolbul.2023.115433.

Ardhani FAK, Safithri M, Tarman K, Husnawati, Setyaningsih I, Meydia (2019) Antioxidant activity of collagen from skin of parang-parang fish (*Chirocentrus dorab*) using DPPH and CUPRAC methods. In: Damar A (ed) IOP conference series: earth and environmental science, Bogor, Indonesia, 241. 012032. IOP Publishing. https://doi.org/10.1088/1755-1315/241/1/012032

Ašmonaitė G, Larsson K, Undeland I, Sturve J, Carney Almroth B (2018) Size matters: ingestion of relatively large microplastics contaminated with environmental pollutants posed little risk for fish health and fillet quality. Environ Sci Technol 52(24):14381–14391. https://doi.org/10.1021/acs.est.8b04849.

Bainy ACD (1996) Oxidative stress as biomarker of polluted aquatic sites. In: Val AL, Almeida-Val VMF, Randall DJ (ed) Physiology and biochemistry of the fishes of the amazon INPA, Manaus, Brazil, 101–110. INPA Publishing.

Baldwin AK, Corsi SR, Mason SA (2016) Plastic debris in 29 Great Lakes tributaries: relations to watershed attributes and hydrology. Environ Sci Technol 50(19):10377–10385. https://doi.org/10.1021/acs.est.6b02917.

Banaee M, Soltanian S, Sureda A, Gholamhosseini A, Haghi BN, Akhlaghi M, Derikvandy A (2019) Evaluation of single and combined effects of cadmium and micro-plastic particles on biochemical and immunological parameters of common carp (*Cyprinus carpio*). Chemosphere 236: 124335. https://doi.org/10.1016/j.chemosphere.2019.07.066.

Barboza LGA, Lopes C, Oliveira P, Bessa F, Otero V, Henriques B, Raimundo J, Caetano M, Vale C, Guilhermino L (2020) Microplastics in wild fish from North East Atlantic Ocean and its potential for causing neurotoxic effects, lipid oxidative damage, and human health risks associated with ingestion exposure. Sci Total Environ 717: 134625. https://doi.org/10.1016/j.scitotenv.2019.134625.

Barboza LGA, Vieira LR, Branco V, Figueiredo N, Carvalho F, Carvalho C, Guilhermino L (2018) Microplastics cause neurotoxicity, oxidative damage and energy-related changes and interact with the bioaccumulation of mercury in the European seabass, *Dicentrarchus labrax* (Linnaeus, 1758). Aquat Toxicol 195:49–57. https://doi.org/10.1016/j.aquatox.2017.12.008.

Besseling E, Redondo-Hasselerharm P, Foekema EM, Koelmans AA (2019) Quantifying ecological risks of aquatic micro-and nanoplastic. Crit Rev Environ Sci Technol 49(1):32–80. https://doi.org/10.1080/10643389.2018.1531688.

Bhat A, Paria A, Deepika A, Sreedharan K, Makesh M, Bedekar MK, Purushothaman CS, Rajendran KV (2015) Molecular cloning, characterisation and expression analysis of melanoma differentiation associated gene 5 (MDA5) of green chromide, *Etroplus suratensis*. Gene 557(2):172–181. https://doi.org/10.1016/j.gene.2014.12.028.

Biller JD, Takahashi LS (2018) Oxidative stress and fish immune system: phagocytosis and leukocyte respiratory burst activity. Acad Bras Ciênc 90:3403–3414. https://doi.org/10.1590/0001-3765201820170730.

Bradford MM (1976) A rapid and sensitive method for the quantitation of microgram quantities of protein utilizing the principle of protein-dye binding. Analyt Biochem 72:248–254. https://doi.org/10.1016/0003-2697(76)90527-3.

Chandrasekar S, Nich T, Tripathi G, Sahu NP, Pal AK, Dasgupta S (2014) Acclimation of brackish water pearl spot (*Etroplus suratensis*) to various salinities: relative changes in abundance of branchial Na+/K+-ATPase and Na+/K+/2Cl− co-transporter in relation to osmoregulatory parameters. Fish Physiol Biochem 40:983–996. https://doi.org/10.1007/s10695-013-9899-y.

Chen F, Li Z, Su Z, Liao H, Gao D, Zhi L, Kong C, Zheng Q, Wang J (2025) Toxic impacts of polystyrene nanoplastics and PCB77 in blunt snout bream: evidence from tissue morphology, oxidative stress and intestinal microbiome. Geosci Front 16: 102005. https://doi.org/10.1016/j.gsf.2025.102005.

Chen Q, Lackmann C, Wang W, Seiler TB, Hollert H, Shi H (2020) Micro-sized microplastics lead to hyperactive swimming behaviour in adult zebrafish. Aquat Toxicol 224: 105521. https://doi.org/10.1016/j.aquatox.2020.105521.

Dai Z, Zhang H, Zhou Q, Tian Y, Chen T, Tu C, Fu C, Luo Y (2018) Occurrence of microplastics in the water column and sediment in an inland sea affected by intensive anthropogenic activities. Environ Pollut 242:1557–1565. https://doi.org/10.1016/j.envpol.2018.07.131.

Daniel DB, Ashraf PM, Thomas SN (2020) Abundance, characteristics and seasonal variation of microplastics in Indian white shrimps (*Fenneropenaeus indicus*) from coastal waters off Cochin, Kerala, India. Sci Total Environ 737: 139839. https://doi.org/10.1016/j.scitotenv.2020.139839.

Das BC, Ramanan PA, Gorakh SS, Pillai D, Jayadradhan RKV (2023) Sub-chronic exposure of *Oreochromis niloticus* to environmentally relevant concentrations of smaller microplastics: accumulation and toxico-physiological responses. J Hazard Mater 458: 131916. https://doi.org/10.1016/j.jhazmat.2023.131916.

De Souza Anselmo C, Sardela VF, de Sousa VP, Pereira HMG (2018) Zebrafish (*Danio rerio*): a valuable tool for predicting the metabolism of xenobiotics in humans?. Comp Biochem Physiol C Toxicol Pharm 212:34–46. https://doi.org/10.1016/j.cbpc.2018.06.005.

Ding J, Huang Y, Liu S, Zhang S, Zou H, Wang Z, Zhu W, Geng J (2020) Toxicological effects of nano-and micro-polystyrene plastics on red tilapia: are larger plastic particles more harmless?. J Hazard Mater 396: 122693. https://doi.org/10.1016/j.jhazmat.2020.122693.

Dobšíková R, Svobodová ZVAF, Blahova J, Modra H, Velíšek J (2009) The effect of transport on biochemical and haematological indices of common carp (*Cyprinus carpio* L.). Czech J Anim Sci 54(11):510–518. https://doi.org/10.17221/52/2009-CJAS.

Duan C (1998) Nutritional and developmental regulation of insulin-like growth factors in fish. J Nutr 128(2):306S–314S. https://doi.org/10.1093/jn/128.2.306s.

Farag MR, Alagawany M, Khalil SR, Moustafa AA, Mahmoud HK, Abdel-Latif HM (2021) *Astragalus membranaceus* polysaccharides modulate growth, hemato-biochemical indices, hepatic antioxidants, and expression of HSP70 and apoptosis-related genes in *Oreochromis niloticus* exposed to sub-lethal thallium toxicity. Fish Shellfish Immunol 118:251–260. https://doi.org/10.1016/j.fsi.2021.09.009.

Greenwell MG, Sherrill J, Clayton LA (2003) Osmoregulation in fish: mechanisms and clinical implications. Vet Clin North Am Exot Anim Pr 6(1):169–189. https://doi.org/10.1016/s1094-9194(02)00021-x.

Griffith MB (2017) Toxicological perspective on the osmoregulation and ionoregulation physiology of major ions by freshwater animals: Teleost fish, crustacea, aquatic insects, and Mollusca.

Environ Toxicol Chem 36(3):576–600. https://doi.org/10.1002/etc.3676.

Guerrera MC, Aragona M, Porcino C, Fazio F, Laurà R, Levanti M, Montalbano G, Germanà G, Abbate F, Germanà A (2021) Micro and nano plastics distribution in fish as model organisms: histopathology, blood response and bioaccumulation in different organs. Appl Sci 11(13):5768. https://doi.org/10.3390/app11135768.

Hamed M, Soliman HA, Osman AG, Sayed AEDH (2019) Assessment the effect of exposure to microplastics in Nile Tilapia (*Oreochromis niloticus*) early juvenile: I. blood biomarkers. Chemosphere 228:345–350. https://doi.org/10.1016/j.chemosphere.2019.04.153.

Hamed M, Soliman HA, Osman AG, Sayed AEDH (2020) Antioxidants and molecular damage in Nile Tilapia (*Oreochromis niloticus*) after exposure to microplastics. ESPR 27:14581–14588. https://doi.org/10.1007/s11356-020-07898-y.

Hamza A, Ijaz MU, Ehsan N, Khan HA, Alkahtani S, Atique U (2023) Hepatoprotective effects of astragalin against polystyrene microplastics induced hepatic damage in male albino rats by modulating Nrf-2/Keap-1 pathway. J Funct Foods 108: 105771. https://doi.org/10.1016/j.jff.2023.105771.

Hu M, Palić D (2020) Micro-and nano-plastics activation of oxidative and inflammatory adverse outcome pathways. Redox Biol 37: 101620. https://doi.org/10.1016/j.redox.2020.101620.

Huang Y, Ding J, Zhang G, Liu S, Zou H, Wang Z, Zhu W, Geng J (2021) Interactive effects of microplastics and selected pharmaceuticals on red tilapia: role of microplastic aging. Sci Total Environ 752: 142256. https://doi.org/10.1016/j.scitotenv.2020.142256.

James K, Vasant K, Padua S, Gopinath V, Abilash KS, Jeyabaskaran R, Babu A, John S (2020) An assessment of microplastics in the ecosystem and selected commercially important fishes off Kochi, south eastern Arabian Sea, India. Mar Pollut Bull 154: 111027. https://doi.org/10.1016/j.marpolbul.2020.111027.

Javed M, Usmani N (2015) Stress response of biomolecules (carbohydrate, protein and lipid profiles) in fish *Channa punctatus* inhabiting river polluted by thermal power plant effluent. Saudi J Biol Sci 22(2):237–242. https://doi.org/10.1016/j.sjbs.2014.09.021.

Kandemir FM, Yildirim S, Kucukler S, Caglayan C, Mahamadu A, Dortbudak MB (2018) Therapeutic efficacy of zingerone against vancomycin-induced oxidative stress, inflammation, apoptosis and aquaporin 1 permeability in rat kidney. Biomed Pharmacother 105:981–991. https://doi.org/10.1016/j.biopha.2018.06.048.

Karami A, Groman DB, Wilson SP, Ismail P, Neela VK (2017) Biomarker responses in zebrafish (*Danio rerio*) larvae exposed to pristine low-density polyethylene fragments. Environ Pollut 223:466–475. https://doi.org/10.1016/j.envpol.2017.01.047.

Katzenberger TD, Thorpe KL (2015) Assessing the impact of exposure to microplastics in fish: evidence report-SC120056. Bristol, 26. Environment Agency Publishing.

Kye H, Kim J, Ju S, Lee J, Lim C, Yoon Y (2023) Microplastics in water systems: a review of their impacts on the environment and their potential hazards. Heliyon 9:e14359. https://doi.org/10.1016/j.heliyon.2023.e14359.

Li F, Li F, Hou X, Luo X, Tu H, Zou Y, Sun C, Shi M, Zheng H (2018) Comparison of six digestion methods on fluorescent intensity and morphology of the fluorescent polystyrene beads. Mar Pollut Bull 131:515–524. https://doi.org/10.1016/j.marpolbul.2018.04.056.

Liang Y, Huang L, Huang K, Zhou Y, Cheng Y, Wu L (2018) Effects of fat, choline and feeding regime on catalase gene expression in the liver of GIFT (*Oreochromis niloticus*). Aquac Res 49(3):1250–1261. https://doi.org/10.1111/are.13579.

Lind PM, Lind L (2020) Are persistent organic pollutants linked to lipid abnormalities, atherosclerosis and cardiovascular disease? a review. J Lipid Atheroscler 9(3):334. https://doi.org/10.12997/jla.2020.9.3.334.

Liu Y, Qiu X, Xu X, Takai Y, Ogawa H, Shimasaki Y, Oshima Y (2021) Uptake and depuration kinetics of microplastics with different polymer types and particle sizes in Japanese medaka (*Oryzias latipes*). Ecotoxicol Environ Saf 212: 112007. https://doi.org/10.1016/j.ecoenv.2021.112007.

Livak KJ, Schmittgen TD (2001) Analysis of relative gene expression data using real-time quantitative PCR and the $2^{-\Delta\Delta CT}$ method. Methods 25(4):402–408. https://doi.org/10.1006/meth.2001.1262.

Lu Y, Ji T, Xu W, Chen D, Gui P, Long F (2024) Rapid, sensitive, and non-destructive on-site quantitative detection of nanoplastics in aquatic environments using laser-backscattered fiber-embedded optofluidic chip. J Hazard Mater 479: 135591. https://doi.org/10.1016/j.jhazmat.2024.135591.

Lu Y, Zhang Y, Deng Y, Jiang W, Zhao Y, Geng J, Ding L, Ren H (2016) Uptake and accumulation of polystyrene microplastics in zebrafish (*Danio rerio*) and toxic effects in liver. Environ Sci Technol 50(7):4054–4060. https://doi.org/10.1021/acs.est.6b00183.

Lusher AL, O'Donnell C, Officer R, O'Connor I (2016) Microplastic interactions with North Atlantic mesopelagic fish. ICES J Mar Sci 73(4):1214–1225. https://doi.org/10.1093/icesjms/fsv241.

Mak CW, Yeung KCF, Chan KM (2019) Acute toxic effects of polyethylene microplastic on adult zebrafish. Ecotoxicol Environ Saf 182: 109442. https://doi.org/10.1016/j.ecoenv.2019.109442.

Marshall WS (2012) Osmoregulation in estuarine and intertidal fishes. Fish Physiol 32:395–434. https://doi.org/10.1016/B978-0-12-396951-4.00008-6.

Misra HP, Fridovich I (1972) The role of superoxide anion in the autoxidation of epinephrine and a simple assay for superoxide dismutase. J Biol Chem 247(10):3170–3175. https://doi.org/10.1016/S0021-9258(19)45228-9.

Munilkumar S, Sundaray JK, Bhatacharya SB, Devi GA (2013) Biology and fisheries of some brackishwater food fishes of India. Advances Fish Res 6:187–157.

Naidu SA, Rao VR, Ramu K (2018) Microplastics in the benthic invertebrates from the coastal waters of Kochi, Southeastern Arabian Sea. Environ Geochem Health 40(4):1377–1383. https://doi.org/10.1007/s10653-017-0062-z.

Nematdoost Haghi B, Banaee M (2017) Effects of micro-plastic particles on paraquat toxicity to common carp (*Cyprinus carpio*): biochemical changes. Int J Environ Sci Technol 14(3):521–530. https://doi.org/10.1007/s13762-016-1171-4.

Nikki R, Jaleel KA, Ragesh S, Shini S, Saha M, Kumar PD (2021) Abundance and characteristics of microplastics in commercially important bottom dwelling finfishes and shellfish of the Vembanad Lake, India. Mar Pollut Bull 172: 112803.

Pang M, Wang Y, Tang Y, Dai J, Tong J, Jin G (2021) Transcriptome sequencing and metabolite analysis reveal the toxic effects of nanoplastics on tilapia after exposure to polystyrene. Environ Pollut 277: 116860. https://doi.org/10.1016/j.envpol.2021.116860.

Qiang L, Cheng J (2021) Exposure to polystyrene microplastics impairs gonads of zebrafish (*Danio rerio*). Chemosphere 263: 128161. https://doi.org/10.1016/j.chemosphere.2020.128161.

Rattan P (1994) Ecobiology of pearl spot (*Etroplus suratensis* Bloch) in Goa waters. Dissertation, Goa University

Rishi KK, Singh J (1982) Karyological studies on five estuarine fishes. Nucleus 25(3):178–180.

Robin RS, Karthik R, Purvaja R, Ganguly D, Anandavelu I, Mugilarasan M, Ramesh R (2020) Holistic assessment of microplastics in various coastal environmental matrices, southwest coast of India. Sci Total Environ 703: 134947. https://doi.org/10.1016/j.scitotenv.2019.134947.

Saha M, Naik A, Desai A, Nanajkar M, Rathore C, Kumar M, Gupta P (2021) Microplastics in seafood as an emerging threat to marine environment: a case study in Goa, west coast of India. Chemosphere 270: 129359. https://doi.org/10.1016/j.chemosphere.2020.129359.

Salerno M, Berlino M, Mangano MC, Sarà G (2021) Microplastics and the functional traits of fishes: a global meta-analysis. Glob Chang Biol 27(12):2645–2655. https://doi.org/10.1111/gcb.15570.

Statista (2023), *Managed and mismanaged microplastic worldwide 2016–2040*, Global microplastic pollution 2016-2040| Statista (2023).

Sruthy S, Ramasamy EV (2017) Microplastic pollution in Vembanad Lake, Kerala, India: the first report of microplastics in lake and estuarine sediments in India. Environ Pollut 222:315–322. https://doi.org/10.1016/j.envpol.2016.12.038.

Sun T, Zhan J, Li F, Ji C, Wu H (2021) Evidence-based meta-analysis of the genotoxicity induced by microplastics in aquatic organisms at environmentally relevant concentrations. Sci Total Environ 783: 147076. https://doi.org/10.1016/j.scitotenv.2021.147076.

Takahara S, Hamilton HB, Neel JV, Kobara TY, Ogura Y, Nishimura ET (1960) Hypocatalasemia: a new genetic carrier state. J Clin Investig 39(4):610–619. https://doi.org/10.1172/jci104075.

Umamaheswari S, Priyadarshinee S, Kadirvelu K, Ramesh M (2021) Polystyrene microplastics induce apoptosis via ROS-mediated p53 signaling pathway in zebrafish. Chem Biol Interact 345: 109550. https://doi.org/10.1016/j.cbi.2021.109550.

Veerasingam S, Ranjani M, Venkatachalapathy R, Bagaev A, Mukhanov V, Litvinyuk D, Verzhevskaia L, Guganathan L, Vethamony P (2020) Microplastics in different environmental compartments in India: analytical methods, distribution, associated contaminants and research needs. TrAC-Trends Anal Chem 133:16071. https://doi.org/10.1016/j.trac.2020.116071.

Wang E, Wang K, Chen D, Wang J, He Y, Long B, Yang L, Yang Q, Geng Y, Huang X, Ouyang P (2015) Evaluation and selection of appropriate reference genes for real-time quantitative PCR analysis of gene expression in Nile tilapia (*Oreochromis niloticus*) during vaccination and infection. Int J Mol Sci 16(5):9998–10015. https://doi.org/10.3390/ijms16059998.

Wang H, Xu J, Yuan Y, Wang Z, Zhang W, Li J (2024) The exploration of joint toxicity and associated mechanisms of primary microplastics and methamphetamine in Zebrafish larvae. Toxics 12(1):64. https://doi.org/10.3390/toxics12010064.

Wang J, Li Y, Lu L, Zheng M, Zhang X, Tian H, Wang W, Ru S (2019) Polystyrene microplastics cause tissue damages, sex-specific reproductive disruption and transgenerational effects in marine medaka (*Oryzias melastigma*). Environ Pollut 254: 113024. https://doi.org/10.1016/j.envpol.2019.113024.

Wang J, Zheng M, Lu L, Li X, Zhang Z, Ru S (2021) Adaptation of life-history traits and trade-offs in marine medaka (*Oryzias melastigma*) after whole life-cycle exposure to polystyrene microplastics. J Hazard Mater 414: 125537. https://doi.org/10.1016/j.jhazmat.2021.125537.

Wang X, Jian S, Zhang S, Wu D, Wang J, Gao M, Sheng J, Hong Y (2022) Enrichment of polystyrene microplastics induces histological damage, oxidative stress, Keap1-Nrf2 signaling pathway-related gene expression in loach juveniles (*Paramisgurnus dabryanus*). Ecotoxicol Environ Saf 237: 113540. https://doi.org/10.1016/j.ecoenv.2022.113540.

Wen B, Jin SR, Chen ZZ, Gao JZ, Liu YN, Liu JH, Feng XS (2018) Single and combined effects of microplastics and cadmium on the cadmium accumulation, antioxidant defence and innate immunity of the discus fish (*Symphysodon aequifasciatus*). Environ Pollut 243:462–471. https://doi.org/10.1016/j.envpol.2018.09.029.

Xiong F, Liu J, Xu K, Huang J, Wang D, Li F, Wang S, Zhang J, Pu Y, Sun R (2023) Microplastics induce neurotoxicity in aquatic animals at environmentally realistic concentrations: a meta-analysis. Environ Pollut 318: 120939. https://doi.org/10.1016/j.envpol.2022.120939.

Xue YH, Jia T, Yang N, Sun ZX, Xu ZY, Wen XL, Feng LS (2022) Transcriptome alterations in zebrafish gill after exposure to different sizes of microplastics. J Environ Sci Health A 57(5):347–356. https://doi.org/10.1080/10934529.2022.2064668.

Yao FC, Gu Y, Jiang T, Wang PF, Song FB, Zhou Z, Sun JL, Luo J (2023) The involvement of oxidative stress mediated endoplasmic reticulum pathway in apoptosis of Golden Pompano (*Trachinotus blochii*) liver under PS-MPs stress. Ecotoxicol Environ Saf 249: 114440. https://doi.org/10.1016/j.ecoenv.2022.114440.

Ye J, Coulouris G, Zaretskaya I, Cutcutache I, Rozen S, Madden TL (2012) Primer-BLAST: a tool to design target-specific primers for polymerase chain reaction. BMC Bioinforma 13:1–11. https://doi.org/10.1186/1471-2105-13-134.

Zhang C, Wang F, Wang Q, Zou J, Zhu J (2023) Species-specific effects of microplastics on juvenile fishes. Front Physiol 14: 1256005. https://doi.org/10.3389/fphys.2023.1256005.

Zhang C, Wang J, Pan Z, Wang S, Zhang L, Wang Q, Ye Q, Zhou A, Xie S, Zeng F, Xu G (2021a) A dosage-effect assessment of acute toxicology tests of microplastic exposure in filter-feeding fish. Fish Shellfish Immunol 113:154–161. https://doi.org/10.1016/j.fsi.2021.04.010.

Zhang X, Chen X, Gao L, Zhang HT, Li J, Ye Y, Zhu QL, Zheng JL, Yan X (2024) Transgenerational effects of microplastics on Nrf2 signaling, GH/IGF, and HPI axis in marine medaka *Oryzias melastigma* under different salinities. Sci Total Environ 906: 167170. https://doi.org/10.1016/j.scitotenv.2023.167170.

Zhang X, Wen K, Ding D, Liu J, Lei Z, Chen X, Ye G, Zhang J, Shen H, Yan C, Dong S (2021b) Size-dependent adverse effects of microplastics on intestinal microbiota and metabolic homeostasis in the marine medaka (*Oryzias melastigma*). Environ Int 151: 106452. https://doi.org/10.1016/j.envint.2021.106452.

Zheng T, Jia R, Cao L, Du J, Gu Z, He Q, Xu P, Yin G (2021) Effects of chronic glyphosate exposure on antioxdative status, metabolism and immune response in tilapia (GIFT, *Oreochromis niloticus*). Comp Biochem Physiol C Toxicol Pharm 239: 108878. https://doi.org/10.1016/j.cbpc.2020.108878.

Zitouni N, Bousserrhine N, Belbekhouche S, Missawi O, Alphonse V, Boughatass I, Banni M (2020) First report on the presence of small microplastics (≤3 μm) in tissue of the commercial fish *Serranus scriba* (Linnaeus. 1758) from Tunisian coasts and associated cellular alterations. Environ Pollut 263: 114576. https://doi.org/10.1016/j.envpol.2020.114576.

Zitouni N, Bousserrhine N, Missawi O, Boughattas I, Ch'evre N, Santos R, Belbekhouche S, Alphonse V, Tisserand F, Balmassiere L, Dos Santos SP (2021) Uptake, tissue distribution and toxicological effects of environmental microplastics in early juvenile fish *Dicentrarchus labrax*. J Hazard Mater 403: 124055. https://doi.org/10.1016/j.jhazmat.2020.124055.

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